



MATHEMATICAL FOUNDATIONS OF MACHINE LEARNING

PROF. PRATHOSH A P

Department of Electrical Communication Engineering
IISc Bangalore

PRE-REQUISITES :

BE/BTech. ME/MTech

Basic course on Probability theory, Linear Algebra.

Should have some background in Python Programming

INTENDED AUDIENCE :

Senior Undergraduates and Graduate Students from EECS disciplines

INDUSTRY SUPPORT :

Most IT companies including Google, Microsoft, Amazon, IBM, Flipkart, Oracle, Infosys, Accenture, GE etc.

COURSE OUTLINE :

This course introduces the mathematical foundations of machine learning, covering risk minimization, density estimation, regularization, and generalization. Students learn classical methods such as linear models, kernel machines, SVMs, decision trees, and ensemble techniques, as well as modern deep learning approaches including MLPs, CNNs, RNNs, and Transformers. Probabilistic models, clustering, PCA, and the EM algorithm are presented to build a solid grounding in unsupervised learning. The course concludes with an introduction to generative models (GANs, VAEs) as a bridge to advanced topics. Emphasis is placed on both theory and practice, with coding assignments connecting math to real-world ML applications.

ABOUT INSTRUCTOR :

Prof. Prathosh A.P. received his Ph.D. from the Indian Institute of Science (IISc), Bangalore in 2015 in the area of temporal data analysis, completing his dissertation just three years after his B.Tech in 2011, with several top-tier publications. He subsequently worked in corporate research labs such as Xerox Research India, Philips Research, and a California-based start-up, focusing on healthcare analytics, where he generated 15 U.S. patents, many of which are commercialized. In 2017, he joined IIT Delhi as an Assistant Professor in Electrical Engineering, teaching and researching machine learning and deep learning. He is currently a faculty member in the Department of Electrical Communication Engineering at IISc Bangalore. His research interests include deep representational learning, cross-domain generalization, and signal processing with applications in vision and speech. He is also co-founder of Cogniable.Tech, a healthcare AI start-up (winner of the Government of India AI Start-up Challenge), and actively collaborates with industry and medical institutions such as AIIMS. Beyond his technical work, he is deeply engaged with Sanskrit and Indian philosophical sciences, and often explores the intersections between AI and philosophy.

COURSE PLAN :

Week 1: Introduction to Supervised/Unsupervised/Generative, Learning via Empirical Risk Minimization

Week 2: Bayes Optimality and Density Estimation via Divergence Minimization

Week 3: Maximum Likelihood and MAP Estimates, Non-Parametric Estimates (Nearest Neighbours and Parzen Window)

Week 4: Linear Models: Linear regression, least squares, Fisher discriminant, Logistic regressions

Week 5: Regularization & Generalization: Bias–variance Decomposition, Ridge regression, Lasso, Probabilistic interpretation of regularization

Week 6: Kernel Machines & SVMs: Maximum margin classifiers, Dual form, KKT conditions, Kernel trick & RKHS intuition.

Week 7: Perceptron, Neural Networks, Gradient-based Optimization, Error Back Propagation

Week 8: Convolutional Neural Networks: Convolution, pooling, receptive fields, CNN architectures, Transfer learning.

Week 9: Sequence Models: RNNs, backpropagation through time, Vanishing/exploding gradients, GRU, LSTMs

Week 10: Attention & Transformers: Attention mechanism, Self-attention vs recurrence, Encoder–decoder Transformers.

Week 11: Ensembles and Decision trees: Bagging & Random Forests, Boosting (AdaBoost, XGBoost).

Week 12:

- Unsupervised Learning & EM: Clustering: k-Means, Gaussian mixtures, EM algorithm, dimensionality reduction and PCA.
- A preview of Generative Models: GANs and VAEs, Diffusion models (high-level only).